



WILEY

Trade associations and collusion among many agents: evidence from physicians

Author(s): Jorge Alé-Chilet and Juan Pablo Atal

Source: *The RAND Journal of Economics*, Vol. 51, No. 4 (Winter 2020), pp. 1197-1221

Published by: Wiley on behalf of RAND Corporation

Stable URL: <https://www.jstor.org/stable/45380752>

Accessed: 02-10-2026 15:02 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

RAND Corporation, Wiley are collaborating with JSTOR to digitize, preserve and extend access to *The RAND Journal of Economics*

Trade associations and collusion among many agents: evidence from physicians

Jorge Alé-Chilet*

and

Juan Pablo Atal**

We study a recent case where most gynecologists in one city formed a trade association to bargain for better rates with insurance companies. After unsuccessful negotiations, the physicians jointly terminated their insurer contracts and set a minimum price. We find that subsequent realized prices coincided with Nash-Bertrand prices, and that the minimum price was barely binding. We show that these actions ensured the association's stability and increased profits. Our findings shed light on the role of trade association in collusion among a large number of heterogeneous agents, and provide insights for the antitrust analysis of trade associations.

1. Introduction

■ Trade associations are prevalent in many professions, including engineering, law, and medicine. The role of these associations as information exchanges and standard setters has been largely acknowledged. However, trade associations also raise antitrust concerns as they may facilitate coordination on prices, establish barriers to entry, or undertake other activities that diminish competition (FTC, 2018; OECD, 2007; Kühn, Matutes, and Moldovanu, 2001). In fact, cases of successful collusion with a large number of agents typically involve the presence of a trade association, especially in differentiated product industries (Symeonidis, 2002; Levenstein and Suslow, 2006).

In this article, we study the strategies of a large number of agents that attempted to coordinate on consumer prices and on vertical negotiations through a trade association. Using detailed

* Bar-Ilan University; jorge.ale-chilet@biu.ac.il.

** University of Pennsylvania; ataljp@econ.upenn.edu.

We are grateful to Marc Rysman (the Editor) and two anonymous referees for very detailed comments that significantly improved the article. We thank Claudio Agostini, Arthur Fishman, Amit Gandhi, Matt Grennan, Paul Eliason, Joe Harrington, Aviv Nevo, Yossi Spiegel, Matt Weinberg, and seminar and conference participants at ESE (2018), LACEA (2018), Bar-Ilan (2019), IIOC (2019), EARIE (2019), KU Leuven (2019), Tel Aviv Coller (2019), and Utah WBEC (2019) for helpful comments and suggestions. We thank Santiago Alé for his help in obtaining the court documents. Ezra Brooks provided excellent research assistance. We gratefully acknowledge Fabián Basso and Alexis Mahana from the Fiscalía Nacional Económica, and María de la Luz Domper and Ignacio Parot from the Competition Tribunal for their time and their insights on the antitrust case. The authors declare that they have no relevant or material financial interests that relate to the research described in this article.

data on prices, sales, and court documents, we provide the first empirical characterization of a trade association's collusive strategies. The setting—a trade association of gynecologists in Chile—mirrors several cases challenged by the FTC in which a large physician group colluded in negotiations with payers.¹

The Association started working in mid-2011 and comprised 90% of the local gynecologists in one city. After 6 months of unsuccessful negotiations with the insurance companies to increase prices, the members of the Association simultaneously terminated their contracts with the insurers and agreed to set their fees above a specified minimum price. As a result of those two measures, all associated physicians became out-of-network providers and out-of-pocket visit prices rose almost 200% on average. The Association operated for 28 months before it was challenged by the National Economic Prosecutor and was ultimately abolished by the Supreme Court for collusive practices.

The rich data available for this case provides a unique opportunity to better understand the emergence and antitrust consequences of trade associations. The goal of our empirical analysis is to provide a comprehensive framework to analyze the gynecologists case through the lens of antitrust. We rationalize the Association's pricing strategy, assess its stability, and evaluate the role of coordination in the failure of the negotiations with insurers.

We start by showing reduced-form evidence that the Association brought about large and unexpected increases in out-of-pocket prices, and that these resulted in increased rates of patient switching across doctors. We then use a demand model to estimate patients' price elasticities for visits. The implied price elasticities serve as an input of a supply-side model that we calibrate to recover the degree of price coordination that better fits the prices set by the Association members.

Our structural analysis shows that the realized prices of the physicians in the Association coincided with Nash–Bertrand prices. Moreover, Nash prices were such that the minimum price was barely binding. Therefore, the Association's activities changed the insurer-provider network structure, but did not result in *supra-competitive* price levels.

In light of our results, we use our estimated model to empirically investigate the incentive compatibility of the Association's strategy. We calculate the counterfactual profits that each associated physician would have made by leaving the Association while everyone else remains in it. We find that these unilateral deviation profits are negative almost for every associated physician. These findings provide evidence of the incentive compatibility of the Association as prices were set at their competitive level, and there were no incentives to deviate with respect to the decision to leave the insurers' network. Therefore, Nash–Bertrand pricing in the out-of-network phase explains the stability of the Association.

Our demand and supply models allow us to investigate the role of the Association in the negotiation process. To this aim, we compare the profits from the decision to leave the network jointly to the counterfactual profits from leaving the network unilaterally. We find that most doctors would have benefited from unilaterally terminating their contracts with the insurers. Yet, the fact that doctors did not leave the network unilaterally is consistent with the case documents showing that the physicians' ultimate goal was to reach better terms with the insurers. In addition, for every doctor, the profits under the Association were substantially larger than those under unilateral contract termination: On average, leaving the insurers jointly would have resulted in profits that are 2.5 times larger than leaving unilaterally. This large gap helps rationalize why the Association triggered the breakdown of the bargaining between doctors and insurance companies. The Association created an outside option with profits large enough that there was no longer surplus created by the agreement with insurers.

Our results have implications for the antitrust analysis of trade associations in the context of vertical negotiations. As mentioned, antitrust offences by physicians associations are quite common. In fact, the FTC has brought more than 35 complaints against associations of independent physicians since 2001, most of which involved a large group of physicians that terminated

¹ Also, the prosecution of this case was heavily based on the treatment of similar cases by the FTC.

their agreement with insurers in the context of a negotiation.² However, the position taken by the FTC in these cases is not without controversy: Recent calls by antitrust practitioners put into question the prosecution of agreements that seek to counteract large imbalances of power in negotiations, including not only physicians in their negotiations with insurers, but also hotels in their negotiations with online platforms and truck owners in their negotiations with transport companies (see, e.g., Mullan and Timan, 2018; ACCC, 2018). Moreover, Alaska, New Jersey, and Texas have passed bills that permit physicians to bargain collectively (Blair and Herndon, 2004). More recently, the DOJ and FTC have explicitly allowed joint negotiations with payers among providers that get together as “Accountable Care Organizations” (ACO) to participate in the Medicare Shared Savings Program, deeming joint negotiations as “reasonably necessary to an ACO’s primary purpose of improving health care delivery” (DOJ and FTC, 2011).³

Our estimates show that the bargaining with insurance companies significantly restrained physician prices in the pre-Association period. Forming the Association helped its members leave a bargaining equilibrium that entailed low profits for them, but did not result in collusive prices. Still, the Association decreased consumer welfare by appropriating part of the surplus that was captured by insurers and their clients through lower visit prices.

Our analysis also provides several insights for our understanding of collusion. A first insight is related to the role of communication. Standard economic models and conventional wisdom predict that price coordination in a large and heterogeneous set of firms is hard (e.g., Motta, 2004; Levenstein and Suslow, 2006). We find evidence for this prediction in that the trade association was not effective in raising prices above competitive levels despite sustained communication. However, coordination was effective to terminate the contracts with insurers, which is arguably something less complex than coordinating on prices. These facts constitute a challenge for theorists in understanding the different roles of communication in cases of collusion.

Second, our results relate to the differences between economic collusion and illegal antitrust practices. Our finding of competitive prices implies absence of price collusion in the economic sense, whereas fixing a minimum price constituted an illegal practice according to antitrust law (see Whinston, 2008, for a discussion).

Finally, the use of a minimum price is consistent with Harrington’s (2016) insight that firms may collude in differentiated-product industries through the use of a minimum price. However, our analysis shows that the level of the minimum price was ineffective to set supra-competitive prices, because it falls below the range of levels theoretically predicted by Harrington.

□ **Related literature.** This article builds on several strands of the literature. We contribute to the empirical literature on explicit collusion, such as Porter (1983), Levenstein (1997), Genesove and Mullin (1998), Genesove and Mullin (2001), Röller and Steen (2006), Asker (2010), Clark and Houde (2013), Igami and Sugaya (2017), and Alé Chilet (2017).⁴ We study a case of collusion among a large number of agents that involved the presence of a trade association. This is a common feature of collusion cases (Symeonidis, 2002; Levenstein and Suslow, 2006). To our

² Modern Healthcare (2009) mentions 30 cases against physician associations in 2001–2009, and Meier, Albert, and Monahan (2018) reviews 6 such cases in 2009–2018. In these cases the FTC generally issued a consent order that prohibits Associations to negotiate on behalf of any physician with any payer, to refuse to deal with any payer, or to have involvement into the individual negotiations between doctors and payers.

³ The high degree of concentration makes the health insurance industry a natural market for the emergence of proposals seeking to counteract insurer bargaining power. Fulton (2017) finds that 57% of health insurance markets were highly concentrated in 2016. The American Medical Association reports that 69% were highly concentrated (American Medical Association, 2011).

⁴ Athey and Bagwell (2001), Athey and Bagwell (2008), and Athey, Bagwell, and Sanchirico (2004) study collusion when prices are observed and firms receive private cost shocks. Harrington and Skrzypacz (2011) characterize an equilibrium where prices and quantities are private information, but firms truthfully report them to a trade organization.

knowledge, ours is the first article that studies empirically the coordinating actions of a trade organization. We also study empirically the role of a minimum price.⁵

Few articles examine collusion on dimensions other than price. An exception is Sullivan (2017) that reports collusion on product characteristics. We analyze collusion on the organization of the vertical network relationship. The vertical structure determined consumers' out-of-pocket expenses, and although the network is related to price setting, it is also different from it. Thus, in our counterfactuals doctors choose a share of copay and a price, which constitute two different decision variables with respect to which doctors maximize profits. We find that coordination on contract termination was more important than price coordination in increasing profits.

Our methodology to study collusive equilibria after the breakdown of negotiations with insurers is due to Bresnahan (1987), Nevo (2001), Ciliberto and Williams (2014), and Miller and Weinberg (2017). We contrast the predictions of different assumptions on the nature of competition with the prices observed during the association period. We find that the prices set by the Association after the breakdown of the negotiations with the insurers are best predicted by Nash–Bertrand prices rather than any positive degree of price coordination.

This article is also related to the literature on the anticompetitive effects of trade associations.⁶ Donovan (1926) and Oliphant (1926), among others, discuss whether trade associations should be subject to antitrust law as a result of several antitrust suits in the 1920s (see also McGahan (1995) and Carnevali (2011)). Symeonidis (2002) documents minimum price agreements in various British manufacturers associations before antitrust legislation.⁷ Levenstein and Suslow (2006) report that collusion among a large number of agents in unconcentrated industries usually arise due to the role of a trade association. Levenstein and Suslow (2011) argue that cartels involving trade associations are more visible, which makes them more likely to attract the scrutiny of antitrust agencies. Yet, at the same time, Levenstein and Suslow find that such cartels are also more stable. Our findings shed light on the reasons for the ob/gyn cartel stability.

Our work is also related to the literature on vertical relations, and in particular to insurer–provider negotiations in health care markets (Gowrisankaran, Nevo, and Town, 2015; Ho and Lee, 2017). Instead of studying agreement outcomes, we document a case where no agreement was reached and study the providers' pricing behavior in this event. Understanding such breakdowns is important because they shed light on the outside options of each side (Lee and Fong, 2013).

Some articles in the antitrust literature on health care document higher prices in areas where physician practices face less competition (Dunn and Shapiro 2014; Baker et al. 2014; Austin and Baker 2015). Our article shows explicitly that physician coordination may lead to higher prices and that bargaining is an important mechanism determining consumer prices for doctor visits.

We also contribute to the literature on the antitrust treatment of bargaining in the context of vertical negotiations. Blair and Herndon (2004), Blair and DePasquale (2011), Campbell (2007, 2008), and Baker, Farrell, and Shapiro (2008) theoretically discuss whether upstream providers negotiating jointly with downstream firms may improve welfare. Interestingly, this theoretical literature is highly motivated by the physician–insurer cases.⁸

A strand of the literature in labor economics studies strikes and labor disputes. Classic references on collective bargaining and strikes are Hicks (1932) and Ashenfelter and Johnson (1969).⁹

⁵ The use of a minimum price is a recurrent but understudied collusive strategy employed by trade associations of physicians and other professionals. Harrington (2016) reports collusion on minimum prices through trade associations for retail travel agents (Bingaman, 1996), specialty physicians (North Texas Specialty Physicians v. Federal Trade Commission, No. 06-60023, 2008 WL 2043040 (5th Cir., May 14, 2008)), and bus operators (Competition Commission of Singapore, 2009).

⁶ Vives (1990) and Kirby (1988) discuss the role of trade associations as information exchanges.

⁷ Symeonidis also mentions that associations allowed individual price setting, especially in differentiated product industries, but maximum discounts to distributors were common.

⁸ Blair and Herndon (2004) refers to the physician–insurer case explicitly, whereas Campbell (2008) and Blair and DePasquale (2011) also motivate their analysis with the physician–insurer case.

⁹ Card (1990), Gu and Kuhn (1998), and Cramton and Tracy (2003) provide theoretical models and empirical evidence. Kennan (1986) and Cramton and Tracy (2003) review this literature.

The main feature that differentiates a trade association from a labor union is that employees are not considered economic entities because they generally act under the supervision and instruction of the employer. Hence, a labor union that purely represents employees is not considered an association for the purposes of antitrust law¹⁰ Moreover, vertical agreements as analyzed in this article involve a pricing game played by physicians as they market their services to patients, which has no parallel in the case of labor unions.

2. Institutional details

□ **The health insurance market in Chile.** The Chilean healthcare system is divided into a public and a private system. The focus of this article on is the private system, which is a regulated health insurance market operated by a group of private insurance companies known collectively as Isapres (“Instituciones de Salud Previsional”). Isapres cover around 17% of the population. The public regime, FONASA, is a pay-as-you-go system financed by monthly contributions deducted from labor income, cost-sharing, and resources from the general government. FONASA covers roughly two thirds of the population (about 11 million people).¹¹

Plans in the private system are comprehensive, stand-alone plans. These plan have two main coverage features: coinsurance rates (one for inpatient care and another for outpatient care) and coverage caps (insurer payment caps).¹² Similarly to private plans in the United States, individuals have access to different types of plans with respect to the provider network. “Preferred-provider” plans are tied to a specific network, although enrollees can use providers outside of their network at a lower reimbursement rate (similar to PPO in the United States). Individuals can also choose—at a higher premium—plans with an unrestricted network of providers. Under these “free choice” plans, coverage is not tied to the use of a particular clinic or health care system, similar to a traditional fee for service indemnity plan in the United States. Companies also offer a small share of “closed network” plans, where enrollees can only use the services of the plan providers or must pay full price (the equivalent of the U.S. HMO).

Healthcare provision is also divided between public and private providers. As is common in healthcare markets, in-network private providers negotiate their rates with insurance companies through bilateral negotiations. On the other hand, out-of-network private providers set their rates privately.¹³ Finally, the public payer (FONASA) reimburses private providers following a pre-determined rate for each procedure and a provider quality rank.¹⁴

□ **The antitrust case.** Gynecologists in Chillán, a city of 175,000 inhabitants in the Ñuble province in southern Chile, legally constituted the Gynecologists’ Association of Ñuble in August 2011.¹⁵ The Association was formed by 26 out of the 29 ob/gyns of Chillán.¹⁶

Upon the creation of the Association, all gynecologists in Ñuble held network contracts with Isapres that were negotiated independently. To improve their bargaining position vis-à-vis the insurance companies, members of the Association started to jointly negotiate their fees with Isapres

¹⁰ (OECD, 2007).

¹¹ A small fraction of the population is insured by seven “closed” private insurance companies, which are available only to workers in certain industries, by special health care systems such as those of the Armed Forces, or do not have any coverage at all (Bitran, Escobar, and Gassibe, 2010).

¹² Every plan assigns the insured a per-service payment cap, and these caps apply to each visit. Coinsurance rates and the insurer payment caps remain constant across visits and do not accumulate over time. For any particular claim, a person pays her coinsurance rate until the amount that the insurance company contributes reaches the cap for that service.

¹³ With the exception of providers vertically integrated with insurance companies.

¹⁴ The provider quality rank goes from 1 to 3 to adjust for quality and complexity differences.

¹⁵ Chile is divided administratively into 15 regions, each subdivided into 54 provinces. The southern region “Biobío” is divided into four provinces: “Concepcion” is the largest province in the Region, followed by “Ñuble,” “Biobío,” and “Arauco.”

¹⁶ One of the associated physicians was not indicted. However, we observe an increase in his visit prices and thus, we consider him part of the collusive group.

in 2011.¹⁷ According to the documents presented in the antitrust case by the National Economic Prosecutor (Fiscalía Nacional Económica, FNE), in April 2011, the head of the Association, Dr. B., started approaching different Isapres and “informed them about the need” of increasing visit rates to CLP 41,000, up from an average price of CLP 14,000 (FNE, 2014).¹⁸ After unsuccessful negotiations with the insurers, Dr. B. called the Association members to meet for the first time in November 2011. The members agreed on a three-pronged plan: (i) canceling the members’ individual contracts with the Isapres; (ii) setting a minimum fee of CLP 25,000 (roughly USD 50) for visits; (ii) naming Dr. B. as the representative in future negotiations with the Isapres.¹⁹ Subsequently, all members of the Association sent termination letters to the Isapres, which would go into effect in January 21, 2012. The Isapres faced customers complaints, but did not accept the physicians’ requests immediately.²⁰

Throughout 2012, the insurance companies contacted either the physicians separately or the Association’s head and offered rates lower than those requested by the Association.²¹ The physicians rejected those private offers (FNE, 2014). Foreseeing these proposals, Dr. B. reminded the Association members that “any information and/or negotiation with the Isapres should be undertaken through its president” (FNE, 2014).²² It was not until March 2013 that a major Isapre accepted (almost fully) the Association’s terms.²³ Yet, that same month the FNE started investigating the Association for antitrust offenses and filed an indictment in October 2013. The Competition Tribunal found the Association members guilty of colluding in 2015. Finally, the Supreme Court ordered the dissolution of the Gynecologists’ Association in 2016.

3. Data

■ We use the two main data sources of the antitrust case. The first source is the insurance companies’ administrative data. It includes visits and surgeries that were registered by the Isapres between January 2009 and March 2013 for gynecologists operating in Chillán and its neighboring provinces.²⁴ Each record contains physician and medical establishment identifiers, a scrambled patient identifier, patient’s province of residence, date, price, out-of-pocket expenditures, and a procedural code. The second dataset contains the receipts issued by the indicted physicians for 2012. It includes patient and physician identifiers and the total amount paid. We use receipts information because many patients did not process their reimbursement with the insurance companies after doctors became out-of-network and hence their visits are not included in the first

¹⁷ Bargaining is frequently given as the main motive for trade associations in similar antitrust cases. For example, a physician association in South Africa claimed that physicians “needed [the association] to balance the strong bargaining power of private hospitals.” (OECD, 2007)

¹⁸ During our period of study, the exchange rate was roughly CLP 500 = USD 1.

¹⁹ This course of action was recorded in an official act, which also stated that the measures were adopted because the members “feel that they are being exploited” by the Isapres, which “have not known how to compensate and value” the physicians’ work. (FNE, 2014).

²⁰ The agreement also included a minimum price for surgeries of 4–4.4 times the FONASA rates. Yet, the agreement was not very effective for surgeries. For this reason, and because surgeries require the presence of a larger medical team, we focus on visits.

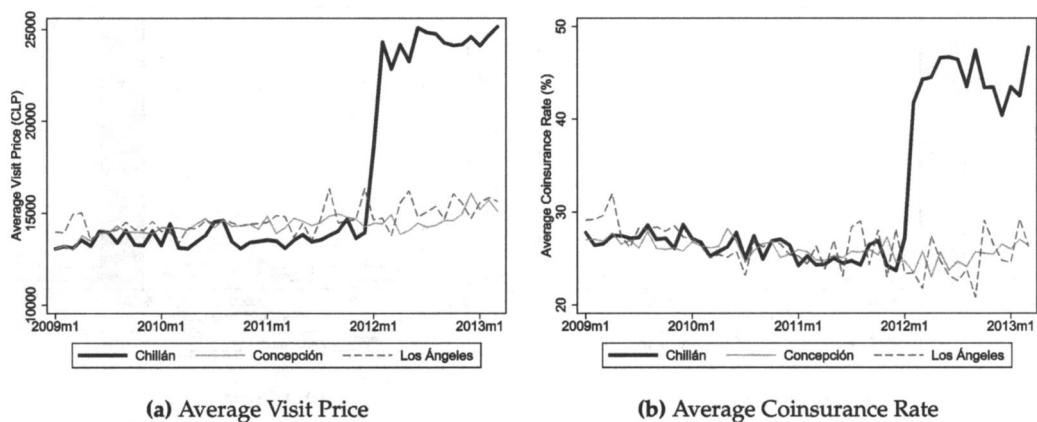
²¹ For example, early on, in January 2012, Dr. B. received an offer extended to all the Association members of a list price of CLP 16,000 (FNE, 2014).

²² The Association statutes included punishments for members who did not comply with these provisions, such as monetary fines and/or expulsion from the Association, although such measures were never implemented. Members complied with the joint negotiation provisions. For instance, one associated physician forwarded to Dr. B. a private offer he received in October 2011, before the Association was officially constituted. Another member who got a private offer replied to the Isapre in June 2012: “I feel that it would only be fair if this offer is extended to all the qualified professionals in Nuble.” (See FNE, 2014.)

²³ The agreed terms were CLP 25,000 for visits and four times the rate paid by FONASA for surgeries. In addition, a small Isapre (closed to public enrollment), reached an agreement with the Association in January 2012 of CLP 20,000 for visits and 4.4 times the FONASA rate for surgeries.

²⁴ There is no entry or exit of doctors during the time period we study.

FIGURE 1
PRICES AND COINSURANCE RATES OVER TIME



Note: The Figure shows the median visit price and coinsurance rate faced by patients of the average gynecologists in Chillán and in two other major neighboring cities (Concepción and Los Ángeles). Prices are in Chilean Pesos (approximately CLP 500 = USD 1).

dataset. These two datasets combined allow us to reconstruct the universe of visits to gynecologists in Chillán and its neighboring cities.²⁵ Given that we estimate aggregate demand at the insurer-physician-month triad, we calculate the price and coinsurance rate as their average over the visits corresponding to each combination. There are 29 different physicians in the data. We drop observations corresponding to two physicians with less than 200 combined visits. Hence, there are 25 associated and 2 non-associated physicians in our final sample.²⁶ Finally, we obtain the physicians' main address from the case documents (FNE, 2013).²⁷

Figure 1 shows the evolution over time of the list price (before reimbursement) and coinsurance rate for the average gynecologist-insurer pair in Chillán as well as in two of its main neighboring cities. Panel (a) shows that the list price in Chillán increased by 75% when the Association formed, from roughly CLP 13,900 in December 2011 to CLP 24,300 in February of 2012. The termination meant not only that the physicians increased their rates, but also that these rates would be covered as out-of-network visits. Panel (b) shows an average increase of 68% on the average coinsurance rate, from 25% to 42%. Combined, the out-of-pocket cost of a visit to the average doctor in Chillán increased by almost 200%. There were no discernible changes in the out-of-pocket costs in neighboring cities.^{28,29}

Compliance within the Association with the minimum visit price was substantial. Figure 2 shows a histogram of each physician's median list price in 2011 and 2012. The dashed line shows the FONASA rate and the dotted line indicates the minimum price set by the Association. The price distribution shifts right in 2012, to the extent that the mode price in 2012 was almost twice

²⁵ We cannot match the two datasets at the patient level. For the demand estimation, we assume that the visits in the receipts database were reimbursed at the out-of-network coverage rate.

²⁶ In addition, we drop the physician-month observations corresponding to four physicians in 2012 that are likely to be the result of incorrect imputation. For example, in the case of one physician two months register 35% of all yearly receipts.

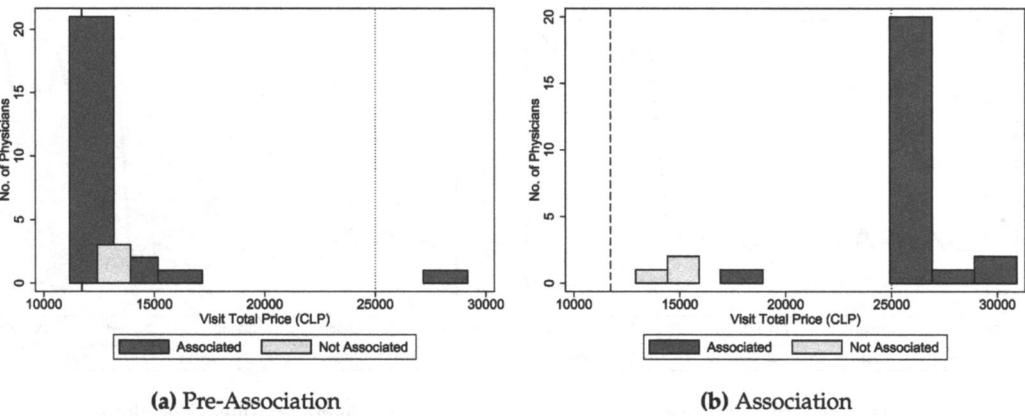
²⁷ For the physicians who were not indicted, we use local websites.

²⁸ Table A1 shows the result of estimating a differences in differences model for prices and coinsurance rates. The estimated price increase is 60% and the estimated increase in the coinsurance rate is 61% of the baseline levels in Chillán. The difference between the increase we report in the text and the estimated effect in the differences-in-differences is due to a slight upward trend in the prices in the control group.

²⁹ As we discuss in the next section, the list price and coinsurance increases were offset by a shift to physicians that did not raise fees. Patients were surprised of the change, which was reported in the local media (FNE, 2013).

FIGURE 2

HISTOGRAM OF MEDIAN PRICES



Note: The Figure shows the median price distribution of Chillán gynecologists in the period before and after the coordinated price increases (2011 and 2012). We drop prices in the transition month of January 2012. Prices are in Chilean Pesos (approximately CLP 500 = USD 1). The dashed line is the FONASA rate and the dotted line is the set minimum price.

as high as the mode price in 2011. That was the case even for the most expensive doctors, who more than doubled their list price. In addition, as we show in Section 4, those physicians who did not raise their rates greatly increased their market share.³⁰

Figure 3 shows the evolution of prices and profits over time for associated and non-associated physicians, as well as for the average doctor. The figure assumes that physicians' marginal cost is equal to the FONASA rate, an assumption that we discuss in more detail in Section 5. The figure shows that concurrently with the price increase among associated physicians, there was a large decrease in their average number of visits, as well as a large increase in the average visits among non-associated doctors.

The profits of associated and non-associated doctors were stable before the Association left the network, and similar across groups. However, profits changed drastically for both groups after the Association left the network; average profits increased four and five times for associated and non-associated doctors, respectively.

Figure 4 shows the evolution of total insurer costs and out-of-pocket expenses (the sum of which corresponds to physician's revenues). The figures shows that higher physician revenues came at the expense of higher costs for both patients and insurers. For insurers, this means that the lower out-of-pocket coverage and any endogenous response in demand did not compensate for the higher list price.

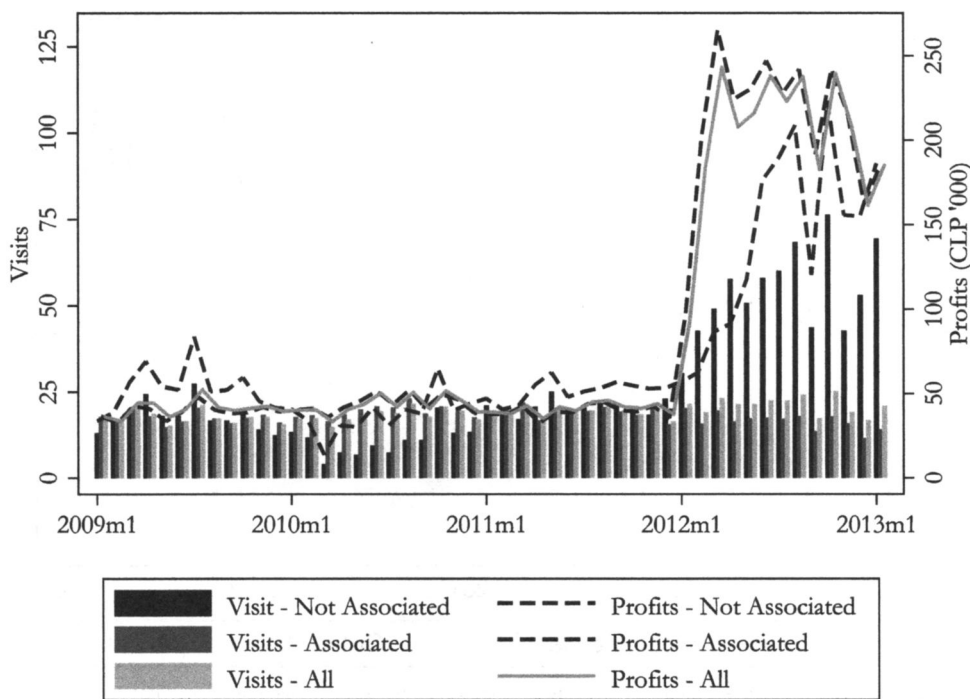
4. Demand

□ **Descriptive evidence on physician switching.** We start by providing evidence of significant demand responses to the large price increases documented in Section 2. In particular, we show that the price increases among colluded doctors lead to a significant shift of patients across doctors, especially from colluding doctors toward non-colluding doctors.

We define a visit of patient p to doctor i to represent a "switch" whenever the doctor seen by p in her previous visit was $i' \neq i$. Panel (a) of Figure 5 shows switching rates (i.e., the share of visits defined as a "switch") over time for Ñuble (the province where doctors colluded) and for a

³⁰ Anecdotally, one of the physicians who did not join the Association was the former government's director of health services in Ñuble and, according to our data, started working in Chillán's private sector only in mid-2010.

FIGURE 3
MONTHLY AVERAGE NUMBER OF VISITS AND PROFITS OVER TIME



Note: The Figure shows the average number of visits and profits (using the FONASA payment as the cost) for all doctors combined and for groups of associated and not associated physicians.

set of nearby provinces that we use as “control” provinces.³¹ The panel shows that switching rates in Ñuble and control provinces had a parallel trend and were stable over time before the Association left the network. However, switching rates increased in Ñuble after February 2012 from 22% to 32%, whereas there was no discernible increase in the switching rates in the control provinces. Panel (b) shows the mean difference in switching between Ñuble and the control provinces for each quarter, and its 95% confidence interval. The panel indicates that the difference between the two groups is statistically significant immediately after the Association left the network.

We formalize the previous description by estimating a difference-in-differences model for switching rates. Let w_{plt} be equal to 1 if the visit by patient p to doctor i in province l and period t corresponds to a switch, and 0 otherwise. We estimate, by OLS, the parameters of the following equation:

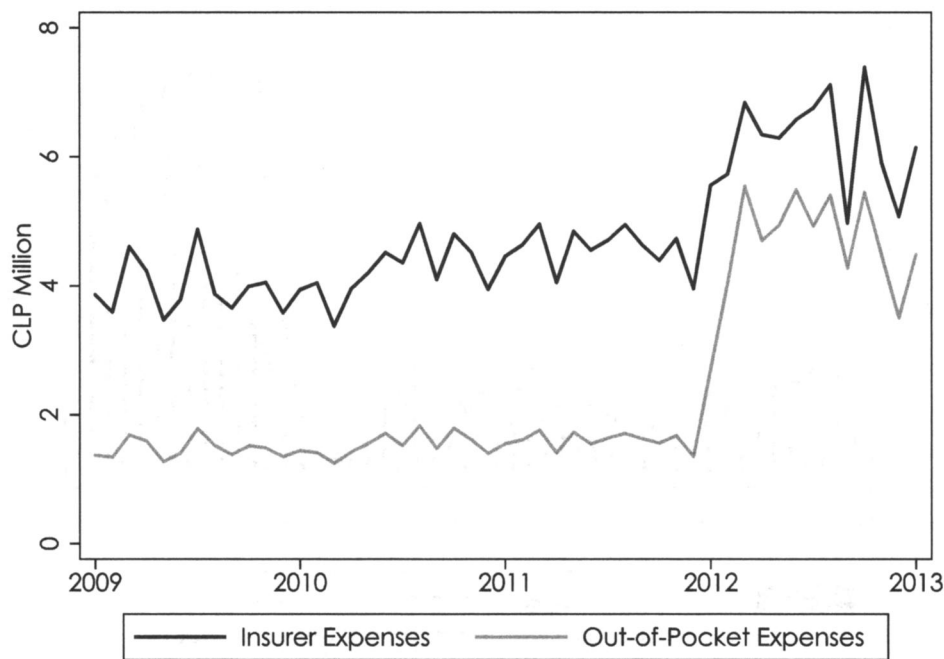
$$w_{plt} = \beta T_l + \gamma T_l \times After_t + g(t) + \epsilon_{ilt}, \tag{1}$$

where T_l is an indicator for Ñuble and $g(t)$ controls for calendar time. We are interested in γ , the estimated effect of the collusion on switching rates in Ñuble.

Table 1 shows the estimation results of equation (1). Column (1) uses $g(t) = \delta After_t$, where $After_t$ is dummy variable indicating dates post February 2012. Column (2) replaces $\delta After_t$ by a full set of year-month fixed effects. In both specifications we find $\hat{\gamma} = 0.07$; a 7 p.p. increase in the switching rate. This increase corresponds to a 32% of the baseline switching rate in Ñuble before the Association left the network. Appendix A shows that this large increase in switching across physicians occurred mostly from colluding physicians toward non-colluding physicians.

³¹ We use Concepción and Biobío as control provinces, although the results are robust to using a different subset of provinces in the region as a control group. See footnote for details.

FIGURE 4
INSURER EXPENSES AND OUT-OF-POCKET EXPENSES



Note: The Figure shows patients’ and insurers’ aggregated expenses of visits. The sum of the two lines represent physicians’ revenues.

□ **Demand model.** We model physicians as differentiated-product firms to allow for idiosyncratic preferences for ob/gyns, which is likely an important feature of the industry. In our context, physicians are different from each other in terms of location, and possibly in terms of their quality, which introduces both horizontal and vertical differentiation.

We assume a nested logit demand model, where the set of physicians is partitioned into non-overlapping nests B_k , $k = 1, \dots, K$.³²

The utility of patient p enrolled in insurer j for visiting doctor i in nest k in period t is

$$u_{pijt} = \delta_{ijt} + \epsilon_{ijt} + \eta_{pkt} + (1 - \sigma)v_{pijt},$$

where δ_{ijt} denotes the mean utility of insurees who visited doctor i at time t , ϵ_{ijt} is the unobserved mean-utility component of physician desirability, η_{pkt} is an unobserved common shock to physicians in nest k , σ is a parameter that measures within-nest correlation, and v_{pijt} is an idiosyncratic shock. The model assumes that $\eta_{pkt} + (1 - \sigma)v_{pijt}$ has a generalized extreme value distribution.

The share of insurees who visit doctor i in time t (among all female insurees in insurance company j), follows the standard share equation:

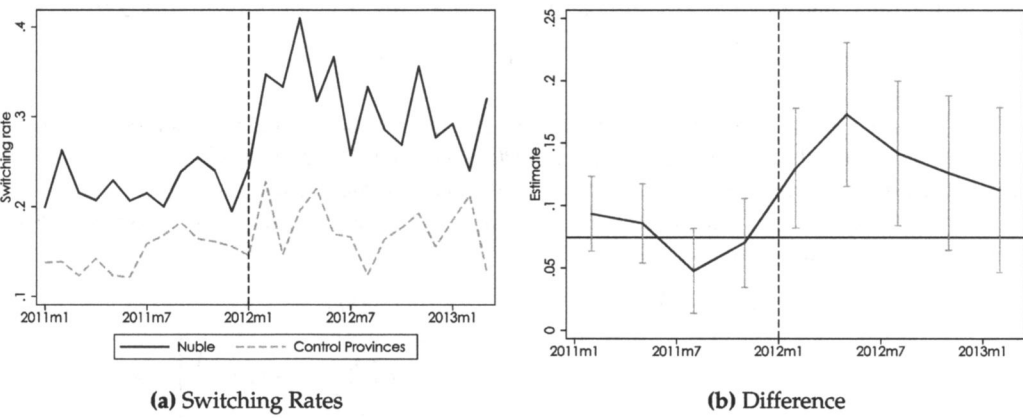
$$s_{ijt} = \frac{\exp(\frac{\delta_{ijt}}{1-\sigma}) \left[\sum_{h \in B_k} \exp(\frac{\delta_{hjt}}{1-\sigma}) \right]^{-\sigma}}{1 + \sum_{l=1}^K \left[\sum_{h \in B_l} \exp(\frac{\delta_{hjt}}{1-\sigma}) \right]^{1-\sigma}}. \tag{2}$$

We parameterize δ_{ijt} as

$$\delta_{ijt} = \alpha p_{ijt}(1 - c_{ijt}) + \mu_{ij} + h(t), \tag{3}$$

³² The nests are determined based on the doctor’s geographic location as explained in detail in Section 4. Using nests captures differentiation across doctors arising from patient’s heterogeneous preferences for location.

FIGURE 5
PHYSICIAN SWITCHING RATES IN ÑUBLE AND SURROUNDING PROVINCES



Note: Panel (a) shows the average switching rates across doctors for OBGyN visits in the province of Ñuble (where collusion occurred) and in surrounding provinces (Biobío and Concepción). Panel (b) shows the quarterly difference in switching rates between the two groups and their 95% confidence intervals. The vertical line marks the implementation date of the collusive measures. The horizontal line in Panel (b) shows the estimates average in 2011.

TABLE 1 Physician Switching Rates

	(1)	(2)
$(After_t) \times T_i$	0.07*** (0.02)	0.07*** (0.02)
T_i	0.08*** (0.01)	0.08*** (0.01)
$After_t$	0.03*** (0.01)	
Year-month FE	N	Y
N	16,742	16,742
R^2	0.02	0.02
Baseline switching	0.22	0.22

Note: The baseline switching rate corresponds to the switching rate in Ñuble before the Association left the network. Heteroscedasticity robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

where p_{ijt} is the list price of a visit to physician i of enrollees of j in period t , $1 - c_{ijt}$ is the coinsurance rate, μ_{ij} represent fixed effects for doctor–insurer pairs, and $h(t)$ is a flexible function of time that captures time-specific common shocks to demand.

□ **Estimation.** The estimation of equation (2) poses three challenges. First, 26% of market-share observations in our data are equal to zero. This fact generates censoring in the estimation of equation (2) because the logit model does not allow for zero shares. Second, we need to specify the nesting structure of the model. Finally, we face the standard endogeneity problem of prices and within shares, as both prices and the within shares are potentially correlated with ϵ_{ijt} . We address these issues below.

Zero shares. As mentioned, many of the doctor shares in the data are zero. However, all doctors are in all insurees’ choice set because insurees can in principle see any given doctor by paying the corresponding out-of-pocket price (see Section 2). Therefore, the observed zero shares s_{ijt}

can be interpreted as the realization of small shares with a finite number of consumers, which differ from the true population probabilities implied by the model.

We rely on Gandhi, Lu, and Shi (2014), who propose an asymptotic correction based on the assumption that sales distribute according to Zipf's Law.³³ In that case, then for every insurer j

$$M_j \frac{s_{ijt}}{1 - s_{ijt}} | N, M_j, s_{0jt} \sim DCM(\theta_j 1_N, M_j(1 - s_{0jt})), \quad (4)$$

where M_j is the market size (assumed to be equal to the number of female insurees in each Isapre), N represents the number of doctors, and $DCM(\cdot)$ is the Dirichlet Multinomial Distribution with the same N parameters θ_j and s_0 represents the share of the outside option.

Denote the digamma function by $\psi(\cdot)$. We invert the shares equation, as in Berry (1994), and take expectations to obtain:

$$\begin{aligned} \psi(\theta_j + M_j s_{ijt}) - \psi(M_j s_{0jt}) &= \alpha p_{ijt}(1 - c_{ijt}) + \mu_{ij} + h(t) + \\ &\sigma \left[\psi(\theta_j + M_j s_{hjt}) - \psi \left(\sum_{h \in k} \theta_j + M_j s_{ijt} \right) \right] + \epsilon_{ijt}, \end{aligned} \quad (5)$$

which is a corrected version of the nested logit demand model that we can estimate. In this equation, the left hand side corresponds to the expectation of the asymptotic dependent variable of the nested logit model, $E[\ln(\pi_{ijt}/s_{0jt})]$, where π_{ijt} denotes the true shares. The term in square brackets is the expectation of the log asymptotic within share of physician i , $E[\ln \pi_{ijt/B_k}] = E[\ln(\pi_{ijt} / \sum_{h \in B_k} \pi_{iht})]$, $j \in B_k$. Our main objects of interest are the parameters α and σ .

Our estimation procedure has two steps. First, we estimate the DCM parameters θ_j of (4) via Maximum Likelihood. Then, we use the resulting estimates to estimate the parameters of equation (5) using a two-stage least squares estimator, that we discuss after we introduce the nesting structure below.

Nesting. Patients' distaste for travel distance is a well-documented feature of demand for medical services (see, e.g., Phelps and Newhouse, 1974). In the absence of data on patient's location, we use the nesting structure to accommodate for the fact that patients are more likely to substitute to a physician that is either in the same medical center or has a practice in a nearby location. In practice, we construct six nests based on the location of each doctor's practice.

Figure A1 of the Appendix shows a map with the location of physicians and the resulting nests. There are three large medical centers where five or more doctors co-locate. We assume that each of those centers constitute a different nest. We group the rest of physicians—who either are a single practice or are co-located with at most two others—into three other nests based on geographic distance.³⁴ Also, we construct two additional nests, one for physicians outside Chillán, and another one for the outside option.

Endogeneity of prices and shares. The estimation of equation (5) by ordinary least squares (OLS) will result in biased estimates in the presence of correlation between the price and within share variables and the unobserved demand shocks ϵ_{ijt} , as is also the case in the standard nested logit model. We identify the own- and cross-price demand elasticities using the large increase in out-of-pocket expenditures that stemmed from the gynecologists' contract termination. In particular, we assume that the emergence and the membership in the Association was not a result of idiosyncratic demand shocks, after controlling for the fixed effects.

Two pieces of evidence support this assumption. First, as shown in Figure 1, prices and coin-surance rates in Chillán had a similar level and followed a similar trend to those in neighboring cities before the Association implemented its measures in Chillán. Thus, the data suggests that

³³ Quan and Williams (2018) also implement the same correction.

³⁴ Prices in the two types of groups are similar. For instance, the median (mean) price before the Association left the network was 13,000 (13,226) and 13,044 (14,459) in these small and large practices, respectively.

TABLE 2 Demand Estimates

	(1) Logit OLS	(2) Logit IV	(3) NL OLS	(4) NL IV
$p_{ijt}(1 - c_{ijt})$	0.004 (0.010)	-0.146*** (0.017)	0.004 (0.004)	-0.050*** (0.018)
ln within-share			1.040*** (0.005)	0.735*** (0.152)
Elasticity in-network period	0.02	-0.60	-0.29	-0.65
Elasticity out-of-network period	0.04	-1.53	-0.75	-1.68
Observations	6620	6620	6620	6620
AR F -stat		39.42		39.42
AP F -stat 1		1899.99		334.84
AP F -stat 2				19.83

Note: The table shows the demand estimates. All specifications include physician-insurer, month-of-the-year, and year fixed effects. Out of pocket expenditures are in thousand CLP. The AR F -stat corresponds to the Anderson–Rubin Wald F -statistic. The AP F -stat 1 and 2 correspond to the first-stage F statistics of the excluded instruments for out-of-pocket expenditure and within shares, respectively (Angrist and Pischke, 2008). Heteroscedasticity robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

the emergence of the Association in Chillán was not a result of particular conditions in this market. In particular, it does not seem that doctor’s bargaining position vis-à-vis insurance companies was substantially lower in Chillán than in other neighboring cities.

Moreover, Figure 3 shows that profits and visits followed a similar trend for associated and non-associated doctors before the Association left the network. Therefore, the data shows that the membership in the Association does not seem to have been driven by doctor-specific shocks to demand.

Under this assumption, the period when the Association is out of the network serves as an instrumental variable (IV) that we use to solve the endogeneity problem. In particular, we use as instruments an indicator for periods after February 2012 and its interaction with a dummy variable for whether physician i joined the agreement.³⁵ Hence, we identify the elasticities using changes in prices and coinsurance rates within the doctor-insurer pair before and after the change in out-of-pocket prices.³⁶

□ **Demand estimation results.** Table 2 presents the results of estimating demand. All specifications include physician-insurer fixed effects, a month-of-the-year fixed effects to control for seasonality, and year fixed effects to control for common trends. We present first the OLS and IV results in columns (1) and (2) of a simpler logit model estimated from equation (5) under the constraint $\sigma = 0$. Columns (3) and (4) present the OLS and IV nested logit estimation results of equation (5).³⁷

In our main specification (column (4) of Table 2) we find $\hat{\alpha} = -0.050(0.018)$ and $\hat{\sigma} = 0.735(0.152)$.³⁸ The Anderson–Rubin and Angrist and Pischke (2008) F -tests reject that the instruments are weak. The rows under the estimates show the average own elasticity in the period before and after the Association left the network.³⁹

³⁵ Given that the post-February 2012 period, which we use as an instrument, has significant overlap with the 2012 year fixed effect, we also tried fixed effects for 8-month periods. The results are unchanged.

³⁶ Our instruments use a supply-side event to identify the demand as in Porter (1983), Eizenberg and Salvo (2015), and Alé Chilet (2017).

³⁷ Online web Appendix A3 presents additional demand estimates resulting from standard ways of correcting for zero market shares, like dropping observations with no sales or replacing the zeros with an arbitrary positive number. See Gandhi, Lu, and Shi (2014) for a discussion on the drawbacks of those methods.

³⁸ The estimate $\sigma \in [0, 1]$, is consistent with utility maximization.

³⁹ Note that the own elasticity for the logit and nested logit models are quite similar; yet, as expected, there is a large difference in the cross elasticities (not reported).

5. The Association's pricing strategy

□ **Model.** We model out-of-network physicians as price-setting, differentiated-product firms that are possibly colluding with their competitors.⁴⁰ This allows us to calculate competitive and collusive prices given the demand estimates.

In every period, each of the N_c colluding physicians decides a unique list price for all insurance companies. Colluding physicians take as given the prices of the $N - N_c$ non-colluding physicians—which are set non-strategically at their in-network level—and the out-of-network coinsurance rates for each insurer. We denote by $\mathbf{p}^N$ the vector with the stacked prices of all the colluding physicians, $\mathbf{c}_j$ the vector of stacked coverage rates (1 minus coinsurance rates) in insurer j and the corresponding vector of monthly visits $\mathbf{q}_j(\mathbf{p}^N, \mathbf{c}_j)$. The first-order condition of colluding physicians can be written in matrix form for each period as

$$\mathbf{p}^N = \mathbf{mc} - \left(\Omega^* * \sum_j E_j * C_j \right)^{-1} \sum_j \mathbf{q}_j(\mathbf{p}, \mathbf{c}_j), \quad (6)$$

where the operator $*$ denotes element-by-element multiplication.⁴¹ The matrix E_j corresponds to the own- and cross-derivatives of demand with respect to out-of-pocket prices in insurer j , which we estimated in Section 4. The matrix C_j represents coinsurance rates, with elements $C_j(i, s) = 1 - c_{ij}$, that converts the list prices into out-of-pocket prices for visits to doctor i among insurees of Isapre j . Finally, the vector $\mathbf{mc}$ is the physicians' marginal cost. We assume that $\mathbf{mc}$ is constant across physicians, and is equal to the FONASA rate, which is determined yearly by the regulator. The rationale for this assumption is that the FONASA rate corresponds to the opportunity cost of seeing one extra privately insured patient.⁴²

We also define the ownership matrix Ω^* for the colluding physicians. The ownership matrix specifies the degree to which each colluding physician internalizes other physicians' profits. As in Nevo (1998), we focus on a general ownership matrix to accommodate different degrees of coordination on prices among colluding physicians, parametrized with a scalar $\kappa \in [0, 1]$ such that

$$\Omega^* = \begin{bmatrix} 1 & \kappa & \dots & \kappa \\ \kappa & 1 & \dots & \kappa \\ \vdots & \vdots & \ddots & \vdots \\ \kappa & \kappa & \dots & 1 \end{bmatrix}. \quad (7)$$

When $\kappa = 0$, the matrix Ω^* becomes the identity matrix, which determines the Nash equilibrium; any $\kappa > 0$ determines a different partial collusive equilibrium, with a higher κ indicating a higher degree of coordination on prices. The extreme of $\kappa = 1$ corresponds to full collusion on prices.⁴³ For any given assumed value of κ , we solve for $\mathbf{p}^N$ by numerically finding the fixed point in equation (6).

Capacity constraints. Physicians see a mix of patients that are either privately or publicly insured. Our supply and demand models assume that capacity constraints for visits of privately insured patients do not bind, so that physicians can always see an extra private patient at the expense of

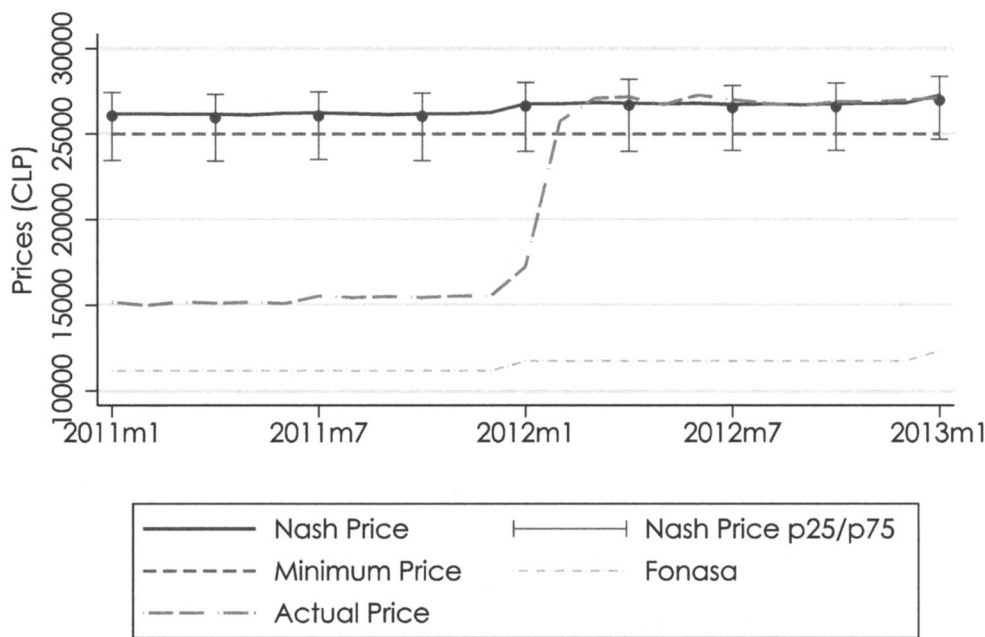
⁴⁰ We do not model bargaining with the insurers during the agreement period because physicians terminated their contracts with the insurers.

⁴¹ We omit time indexes to simplify notation.

⁴² We are making the implicit assumption that doctors can always substitute a private patient with a public patient. This assumption is justified by the well-documented lack of specialist doctors and long waiting times in the public system. As an example, in 2017, there were almost 70,000 individuals nationwide on the waiting list to visit an ob/gyn, among which almost 50,000 had been waiting more than 4 months (INDH, 2018).

⁴³ We refer to Miller and Weinberg (2017) for the discussion of whether κ can be interpreted as a conduct parameter. See the references there, especially Corts (1999) and Sullivan (2017).

FIGURE 6
SIMULATED AND ACTUAL PRICES



Note: The Figure shows the average Nash price for out-of-network coinsurance rates, the average actual price, and the FONASA rate. The dashed line shows the minimum price set by the Association. In addition, the vertical lines show the 25th and 75th percentiles of the out-of-network Nash prices, and the dot indicates the median.

not seeing a public patient. Binding capacity constraints in the data would imply that we underestimate the price coefficient in the demand model. Moreover, binding capacity constraints could generate corner solutions in the supply model, and reduce the potential deviation profits in the counterfactuals we study below (Staiger and Wolak, 1992). We provide evidence supporting the absence of binding capacity constraints for private visits, both in the data and in the predictions of the model, in Appendix A5.

□ **Results.** The main results of the predicted price from equation (6) are in Figure 6. The figure presents the time series of equilibrium prices assuming Nash–Bertrand competition among associated doctors ($\kappa = 0$).⁴⁴ We plot (quantity-weighted) average Nash prices, as well as the 25th and 75th percentiles of the distribution of prices across colluding physicians. For comparison, the figure also includes the average observed prices, the marginal cost (the FONASA rate), and the minimum price set by the Association (CLP 25,000). In addition, Figure 7 shows the distribution of average Nash prices of each physician during the Association period.

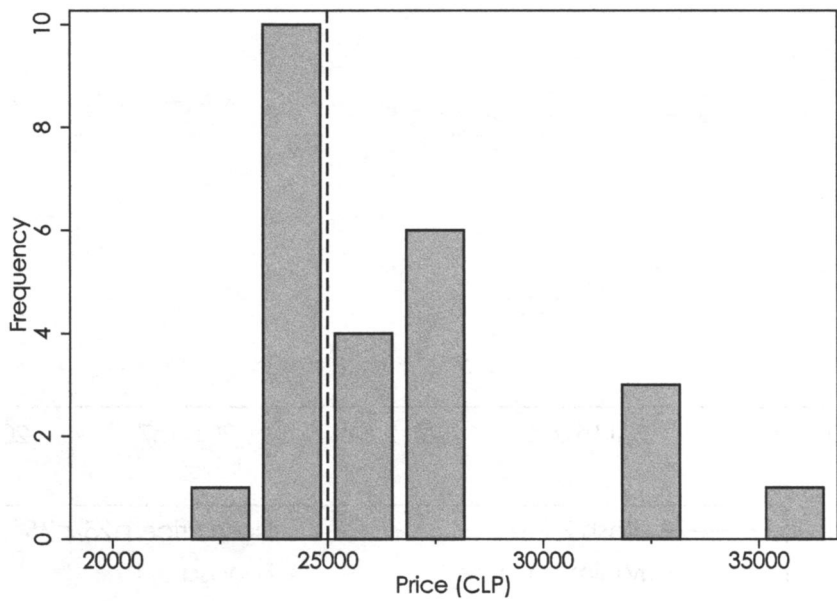
Three facts about our results are particularly noteworthy. First, as shown in Figure 6, the average Nash–Bertrand price (CLP 26,800) is almost twice as high as the observed price in the pre-agreement period. This difference reflects the low bargaining power of the doctors vis-à-vis insurance companies in the pre-Association period.

Figure 6 also shows that the average Nash–Bertrand price is almost equal to average price set by the members of the Association. By contrast, the collusive price (when $\kappa = 1$) is equal to CLP 52,336; twice as high as the average observed price. Panel (a) of Figure 8 shows the ratio between the (quantity-weighted) average predicted price and (quantity-weighted) average observed price.

⁴⁴ Because doctors left the network, we use out-of-network coinsurance rates, which we calculate as the average coinsurance rate of each insurer-physician pair across out-of-network visits in the Association period.

FIGURE 7

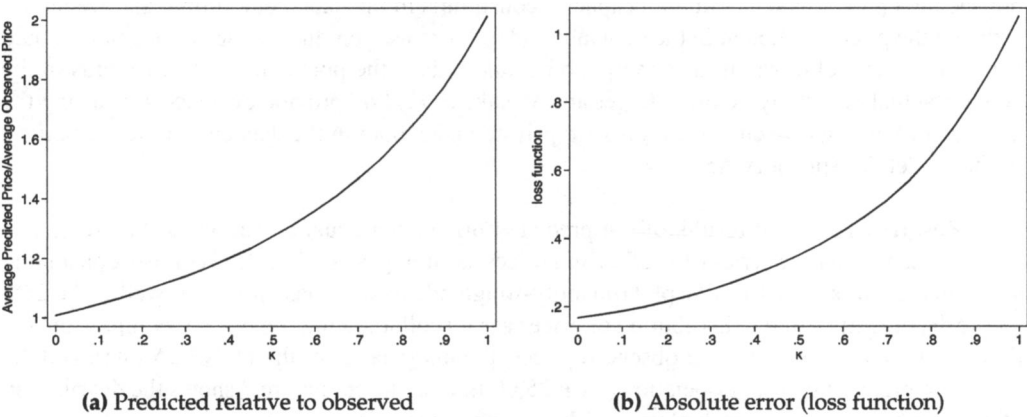
DISTRIBUTION OF NASH-BERTRAND PRICES



Note: The Figure plots the histogram of the estimated Nash prices of each associated physician, averaged over the Association period. The dashed line shows the Association’s minimum price.

FIGURE 8

EQUILIBRIUM PRICES AND FIT AS A FUNCTION OF κ



Note: Panel (a) shows the ratio between the sales-weighted average price predicted by the supply model and the sales-weighted average observed price in the post-association period. Panel (b) shows the “loss function” $f(\kappa) \equiv$

$$\frac{1}{T} \sum_{t \geq t_0} \frac{\frac{1}{N} \sum_i \left| 1 - \frac{P_t^N(\kappa)}{P_t} \right| q_{ijt}(\kappa)}{\sum_i q_{ijt}(\kappa)},$$

Also, $\kappa = 0$ minimizes the difference between the prices predicted by the supply model and the true prices. Panel (b) of Figure 8 shows the (quantity-weighted) average of absolute differences between the predicted and observed prices is increasing in κ , formally defined as

$$f(\kappa) \equiv \frac{1}{T} \sum_{t \geq t_0} \frac{\frac{1}{N} \sum_i \left| 1 - \frac{P_t^N(\kappa)}{P_t} \right| q_{ijt}(\kappa)}{\sum_i q_{ijt}(\kappa)},$$

where t_0 is the date when the Association left the network, T is the number of months the Association was out-of-network. P_{it} is the observed price and $P_{it}^N(\kappa)$ and $q_{ijt}(\kappa)$ are the predicted prices and quantities as a function of κ . As shown in Panel (b) of Figure 8, $f(\kappa)$ is increasing in κ so that the no-coordination solution provides the best fit to the data.⁴⁵

Third, as shown in Figure 7, the distribution of Nash–Bertrand prices is such that the minimum price barely binds.

Our empirical results provide two main insights. First, the Association served as a collusive device on the physicians' decision to become out-of-network providers, but not to set *supra-competitive* prices. Thus, the price increase during collusion stems from the fact that the physicians were no longer constrained by their previous agreements with the insurers. Second, our findings allow us to assess the role of the minimum price as an effective collusive mechanism. In particular, Harrington (2016) shows that the use of a minimum price is an incentive compatible way of achieving *supra-competitive* prices for heterogeneous firms. Yet, our results show that the level chosen for the minimum price was not binding in the Nash equilibrium. Therefore, the minimum price was not helpful to raise profits above competitive levels. Hence, we interpret the use of a minimum price as a focal price, which helped doctors coordinate into switching to the competitive equilibrium.

Robustness to the marginal cost assumption. An important assumption in our empirical analysis is the level of the marginal cost, which we assumed to be equal to the FONASA rate. However, one could argue that a visit of a privately insured patient is more costly than a visit of a publicly insured patient if, for instance, doctors spend more time with the privately-insured patients. In that case, the FONASA rate (CLP 10,970) would correspond to a lower bound for the marginal cost. On the other hand, the actual price before the Association left the network (CLP 13,982 on average) is an upper bound on the marginal cost, as a higher marginal cost would violate doctors' participation constraint in the agreement with insurers. In the extreme case that the price before the Association left the network was equal to the doctor's marginal cost—a situation that price leaves doctors just indifferent between participating in the market or not—we would underestimate the equilibrium prices by CLP 3000. Still, the conclusion about lack of price coordination is unaffected by assuming a higher marginal cost.

6. Stability of the Association

■ In this section we evaluate the stability of the Association by analyzing whether colluding physicians would have profited from leaving the Association unilaterally. In fact, insurers approached physicians individually during the Association period and offered them to rejoin their network. Yet, the Association prevailed and no physician deviated. Finding a large number of physicians with positive deviation profits would entail that the Association was not stable or depended critically on non-monetary incentives. On the contrary, finding negative deviation profits provides support to our model as a characterization of a stable collusive agreement.

We evaluate the decision to stay in the Association under the assumption that prices out-of-network are set in the Nash–Bertrand game, which is consistent with our findings in the previous section. We assume that deviating physicians set their price at their in-network average, that is, we assume they return to their original insurer contract.⁴⁶

Formally, for a set S of doctors in the Association, the vector of prices $p(S)$ has element i defined by:

$$p_i(S) = \begin{cases} p_i^d, & \text{if } i \notin S \\ p_i^N, & \text{if } i \in S \end{cases}$$

⁴⁵ $f(0) = 0.17\%$, that is, our prediction error is approximately 17% of the actual price.

⁴⁶ The court documents show that one insurer offered as much as 16,000 CLP, only 15% higher than our assumption. We calculated deviation profits at this higher rate, and our conclusions from the analysis are robust: we find that only four physicians would have had incentives to deviate at this higher rate.

where p_i^N is the i th element of the vector defined implicitly by equation (6) for doctors in S , and p_i^d is the average of in-network prices of doctor i .

Similarly, the vector of coinsurance rates ($\mathbf{c}_j$) has element i defined by:

$$c_{i,j}(S) = \begin{cases} c_{i,j}^d, & \text{if } i \notin S \\ c_{i,j}^o, & \text{if } i \in S \end{cases}$$

where $c_{i,j}^d$ and $c_{i,j}^o$ denote the in-network and out-of-network coinsurance rates, respectively

The profits for physician i under S depends on physician i 's price and coinsurance rate as well as all other physician's prices and coinsurance rates, denoted by $\mathbf{p}_{-i}(S)$ and $\mathbf{c}_{-i,j}(S)$:

$$\pi_i(S) = \sum_j (p_i(S) - mc) q_j(p_i(S), \mathbf{p}_{-i}(S), c_{i,j}(S), \mathbf{c}_{-i,j}(S)).$$

When physician i deviates from the Association, her deviation profits are

$$\begin{aligned} \pi_i(S \setminus i) - \pi_i(S) &= \sum_j (p_i^d - mc) q_j(p_i^N, \mathbf{p}_{-i}(S), c_{i,j}^d, \mathbf{c}_{-i,j}(S)) \\ &\quad - \sum_j (p_i^N - mc) q_j(p_i^d, \mathbf{p}_{-i}(S), c_{i,j}^o, \mathbf{c}_{-i,j}(S)) \end{aligned} \quad (8)$$

When i deviates, she lowers the price from p_i^N to p_i^d , but also profits from a higher coverage rate for her patients. These two effects can be decomposed by re-writing equation (8) as

$$\begin{aligned} \pi_i(S \setminus i) - \pi_i(S) &= \sum_j (p_i^d - mc) q_j(p_i^d, \mathbf{p}_{-i}(S), c_{i,j}^o, \mathbf{c}_{-i,j}(S)) - \sum_j (p_i^N - mc) q_j(p_i^N, \mathbf{p}_{-i}(S), c_{i,j}^o, \mathbf{c}_{-i,j}(S)) \\ &\quad + \sum_j (p_i^d - mc) (q_j(p_i^d, \mathbf{p}_{-i}(S), c_{i,j}^d, \mathbf{c}_{-i,j}(S)) - q_j(p_i^d, \mathbf{p}_{-i}(S), c_{i,j}^o, \mathbf{c}_{-i,j}(S))) \end{aligned} \quad (9)$$

The first term of equation (9) is the change in profits due to price deviation while keeping coverage rates at the out-of-network level. As prices are set at the Nash–Bertrand level for out-of-network coverage, the first term in equation (9) is negative. On the other hand, the second term is the deviation toward being outside of the network. This term is positive, and proportional to the extra visits generated by higher coverage in-network.⁴⁷ Therefore, deviation profits have an ambiguous sign, so we quantify them empirically using our demand estimates.

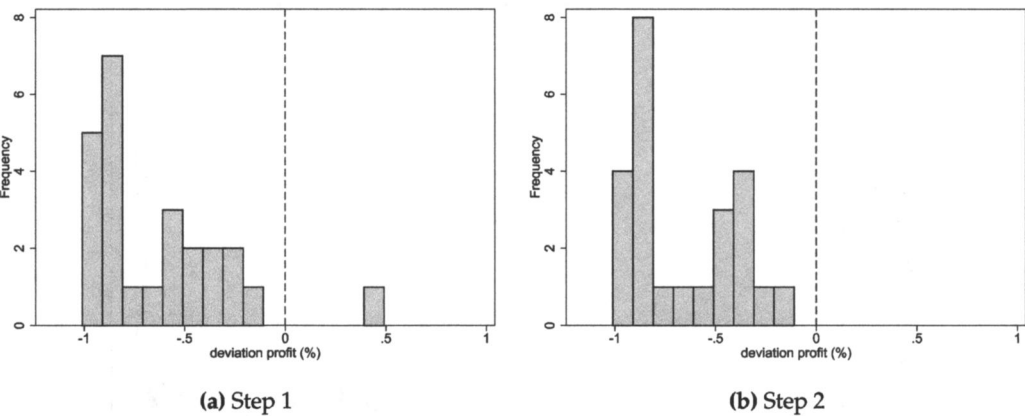
Panel (a) of Figure 9 presents the results. The panel shows the histogram of deviation profits across physicians, calculated as a share of the collusive profits, $(\pi_i(S) - \pi_i(S \setminus i))/\pi_i(S)$. Each unit of observation corresponds to a different deviating physician (and, therefore, to a different counterfactual scenario) among the 25 colluding physicians in our sample.

We find positive deviation profits for only one of the 25 colluding physicians. The median doctor would have lost 79% of her collusion profits by deviating to their pre-agreement price and staying in-network.

In order to assess the overall stability of the Association, we iterate this process by removing the doctor with positive deviation profits and recalculating the new equilibrium prices and profits from unilateral deviations until no doctors have positive deviation profits. This process stops in the second iteration, in the counterfactual situation where the doctor with positive deviations profits leave the Association as it was actually formed. Panel (b) of Figure 9 shows the distribution of deviation profits after the doctor with positive deviation profits in Panel (a) deviates. In this case, none of the remaining doctors have incentives to deviate.

⁴⁷ It is easy to show that $q_j(p_i^d, \mathbf{p}_{-i}(S), c_{i,j}^d, \mathbf{c}_{-i,j}(S)) - q_j(p_i^d, \mathbf{p}_{-i}(S), c_{i,j}^o, \mathbf{c}_{-i,j}(S)) \simeq -E_{ij} \times p_i^d \times (c_{i,j}^d - c_{i,j}^o) > 0$ where E_{ij} is the own-price elasticity of demand for doctor i among enrollees of j with respect to the out-of-pocket price.

FIGURE 9
DEVIATION PROFITS AND ASSOCIATION STABILITY



The Figure shows the distribution of per-period deviation profits across physicians. Each point in the figure corresponds to the deviation profit of a different physician, as a percentage of the profit from staying in the Association.

These findings provide evidence of the incentive compatibility of the Association, as prices were set at their competitive level and there were no incentives to deviate with respect to the decision to leave the insurers’ network.

Incentives to join the Association. An alternative interpretation of equation (9) is that it quantifies physician i ’s incentives to join the Association under the assumption that everyone else joins and charges prices $\mathbf{p}_{-i}(S)$. However, it might be natural to assume that each physician considers her impact on the equilibrium premiums when considering whether to join the Association or not. In that case, each physician i computes her profits from not joining, $\tilde{\pi}(S \setminus i)$, that depend on the outcome of the Nash–Bertrand pricing game when physician i does not join, $\mathbf{p}_{-i}(S \setminus i)$.

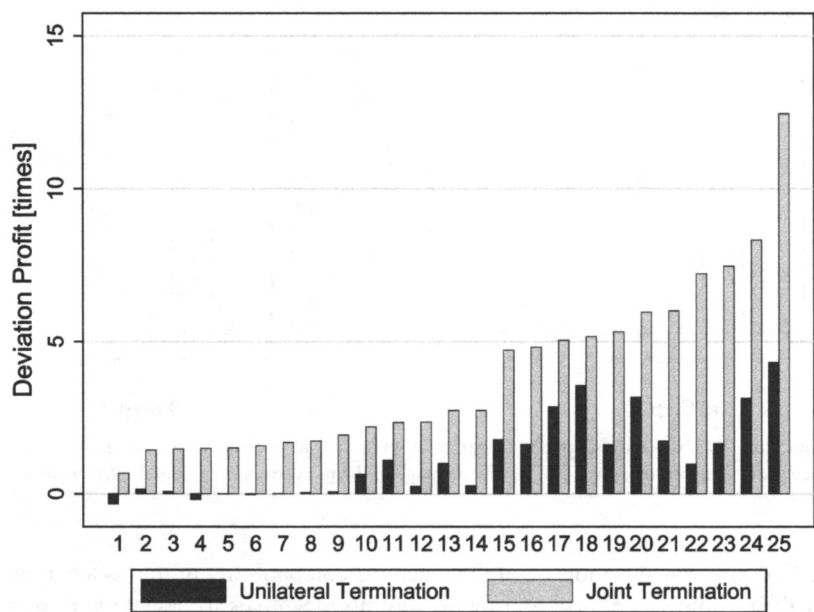
We show the resulting deviation profits after considering endogenous repricing of the associated physicians in Appendix B. We find that price adjustments are small so that deviation profits in this case are similar to the results of Figure 9. We interpret this finding as showing that joining the Association was incentive compatible.

□ **The role of coordination.** In this section, we calculate every doctor’s profits in the counterfactual scenario where she would have unilaterally left the network. In this exercise, the focal doctor i sets her visit price according to the first order condition in equation (6), while the others keep their prices at their individual pre-agreement average. In terms of the notation introduced in Section 6, this results in profits $\pi_i(\{i\})$ for the focal doctor who leaves the network unilaterally. We calculate deviation profits relative to $\pi_i(\emptyset)$, that is, the profits in the case were no doctor leaves the network.

Figure 10 plots the relative profits for every doctor from unilaterally becoming out-of-network $\frac{\pi_i(\{i\}) - \pi_i(\emptyset)}{\pi_i(\emptyset)}$, as well as the relative profits from jointly becoming out-of-network as in the Association $\frac{\pi_i(S) - \pi_i(\emptyset)}{\pi_i(\emptyset)}$.

The figure shows that most physicians would have had higher profits from leaving the network and setting their optimal price (assuming that all other doctors would have stayed in the network) relative to staying in-network. However, for several physicians the gains from deviating were small. In fact, one third of physicians would have increased their profits by less than 10% by deviating unilaterally. The figure also shows that the Association provides much higher profits than the in-network *status quo* or than unilateral deviations for every doctor. On average,

FIGURE 10
DEVIATION PROFITS AS A SHARE OF IN-NETWORK PROFITS



Note: The Figure plots the gains from leaving the network under the assumption that all doctors of the Association leave the network (Joint Termination) and under the assumption that only the focal doctor becomes out-of-network (Unilateral Termination). Deviation profits are calculated as the difference in profits between each scenario and the profits under the situation that all doctors stay in-network.

joint deviation produced profits 400% higher than the status quo, and 140% higher profits than unilateral deviation.⁴⁸

Two main insights emerge from this result. The fact that doctors did not leave the network unilaterally is consistent with the case documents showing that the physicians’ goal was to reach ultimately better terms with the insurers, and that coordinated action was used to increase their bargaining position. In turn, coordinated action raised their outside option to the point where the negotiation broke down. The facts that unilateral deviations did not occur despite being incentive compatible, and that coordination entailed higher profits suggest that communication played an important role in the breakdown of negotiations.

7. Conclusion

■ Collusion among a large number of agents through a trade association is a common phenomenon. In this article we study the collusive strategies of a trade association of physicians in the context of a vertical negotiation with payers. The case closely mirror several cases challenged by the FTC, in spite of recent calls by practitioners that put into question the prosecution of agreements that seek to counteract large imbalances of power in negotiations. The Association members undertook two coordinating strategies: joint termination of their contracts with insurance companies and agreement on a minimum price per visit. These joint measures were

⁴⁸ A fixed cost for leaving the network—for example, due to changing the billing system—could rationalize that termination only occurs when it is made jointly. For instance, a fixed cost of roughly 970 USD would predict that, after 1 year of leaving, only 7 out of the 25 would find it profitable to deviate unilaterally but that 23 out of the 25 would find it profitable to deviate jointly. We note that this calculation is only approximate as it uses, for the profits from joint deviation, the case when *everyone* leaves the insurer’s network.

effective in increasing the members' profits. We find that the realized prices coincided with the competitive price and that the minimum price was barely binding. These findings suggest that the change in the structure of the physician-insurer relationship was more important to increase physicians' profits than price coordination.

The counterfactual exercises allow us to understand the stability and the coordination role of the association. First, we find that unilateral deviation profits from joint contract termination are much lower than the collusive profits. This result indicates that the Association's collusive strategy was incentive compatible. Second, we find that physicians would have benefited from leaving insurers unilaterally, but that they benefited much more from leaving insurers jointly. The fact that, in reality, physicians left insurers jointly is consistent with the physicians wanting to reach ultimately better terms with the insurers and with coordinated action being used to increase their bargaining position vis-à-vis the insurers. These results suggest that coordination played a key role in the price increases of 2012.

Our work has antitrust implications. Some antitrust practitioners have argued that small downstream firms should be allowed to jointly negotiate with larger upstream firms in order to "countervail market power," as most of the surplus is extracted by the large firm. We show that in the case we study joint negotiation caused large price hikes to consumers. This finding provides empirical support to the claim that such joint negotiations should be disallowed when the policymaker's goal is to maximize consumer surplus.

In addition, the Gynecologists' Association did not manage to raise prices much above the competitive outcome despite sustained coordination. However, the Association was highly successful in coordinating the bargaining efforts vis-à-vis the insurers. This result suggests that the nature of coordination in negotiations in vertical relationships is different from that of coordination in prices. Even if it is well known that price collusion among a large number of heterogeneous agents is difficult, the case studied in this article suggests that such hurdles might not arise in other types of coordination.

Furthermore, the case studied in our article highlights the role of communication through a trade association in cases of successful coordination. Although the presence of communication is central in antitrust practice, its role in collusion is not well understood by theoretical models.

Finally, the failed negotiation documented in this article constitutes a departure from the agreement equilibria usually studied in the empirical industrial organization literature. We interpret our findings as showing that the Association increased the profits of doctors' outside option in the bargaining game so that there was no longer surplus created by the agreement with insurers. This fact ultimately resulted in a breakdown of the physician-insurers negotiation. Documenting such breakdowns is important because they shed light on the outside options of each side, and provide theoretical support for the modeling of equilibrium bargaining outcomes. Understanding the dynamics of a failed bargaining process is a promising area for future research.

Appendix A: Market Share Changes

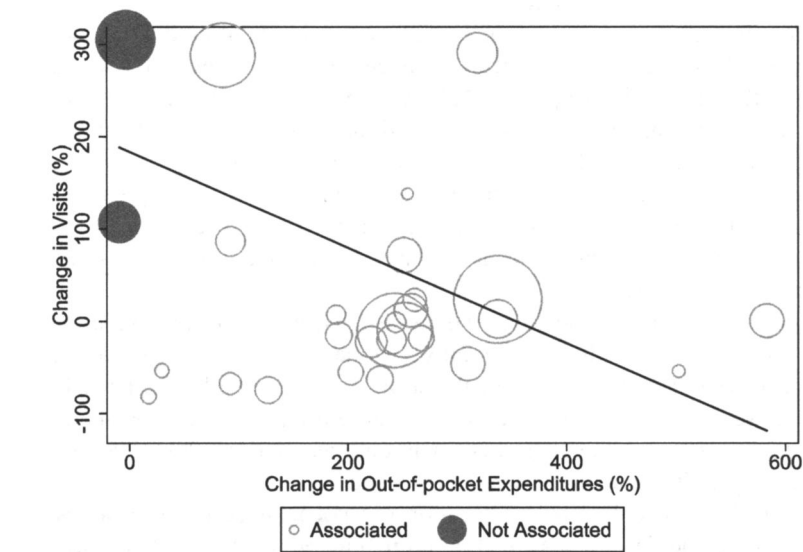
Figure A1 plots the percentage changes in out-of-pocket prices against percentage changes in visits in the pre and post period (weighted by total number of visits in the period). We find that the two non-colluding doctors in our sample increased their visits by 206% on average, whereas the median colluding doctor decreased their visits by 12%.

Appendix B: Incentives to Join

Figure A2 shows the resulting deviation profits after considering endogenous repricing, discussed in the last paragraph of Section 6. With endogenous repricing, we find that only two out of the 25 physicians had incentives to remain out-of-network and not join the Association. This result mirrors the results of Figure 9 in the main text (where no endogenous repricing is allowed) because the change in the Associated physicians' prices when one single physician leaves the Association is small.

FIGURE A1

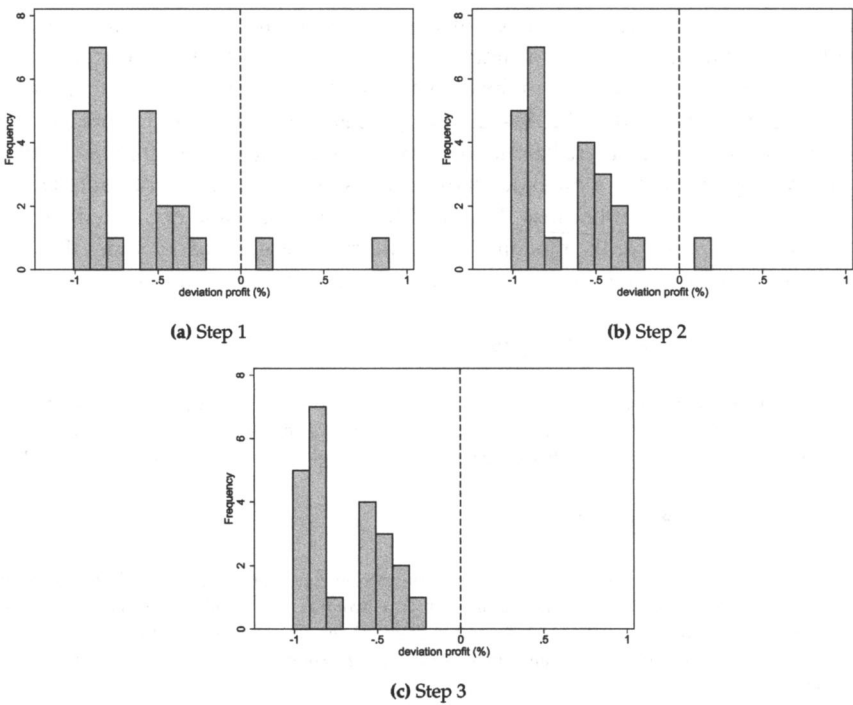
CHANGES IN MARKET SHARE AND CHANGES IN OUT-OF-POCKET PRICES



Note: The Figure plots the percentage changes in out-of-pocket prices against percentage changes in visits in 2012 with respect to 2011 for every physician, and a linear fit. The size of the markers correspond to the physician's total number of visits in the period, which are also the linear fit weights.

FIGURE A2

DEVIATION PROFITS AND ASSOCIATION STABILITY



The Figure shows the distribution of per-period deviation profits across physicians. Each point in the figure corresponds to the deviation profit of a different physician, as a percentage of the profit from staying in the Association.

References

- ACCC. "Potential ACCC Class Exemption for Collective Bargaining." Discussion Paper, Australian Competition and Consumer Commission, 2018.
- ALÉ CHILET, J. "Gradually Rebuilding a Relationship: The Emergence of Collusion in Retail Pharmacies in Chile." Working Paper, University of Pennsylvania, Philadelphia, PA 2017.
- AMERICAN MEDICAL ASSOCIATION. "Competition in Health Insurance: A Comprehensive Study of U.S. Markets, 2017 Update." Technical report, American Medical Association, Chicago, IL, 2011.
- ANGRIST, J.D. and PISCHKE, J.S. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, NJ: Princeton University Press (2008).
- ASHENFELTER, O. and JOHNSON, G. "Bargaining Theory, Trade Unions, and Industrial Strike Activity." *American Economic Review*, Vol. 59(1) (1969), pp. 35–49.
- ASKER, J. "A Study of the Internal Organization of a Bidding Cartel." *American Economic Review*, Vol. 100 (2010), pp. 724–62.
- ATHEY, S. and BAGWELL, K. "Optimal Collusion with Private Information." *RAND Journal of Economics*, (2001), pp. 428–465.
- . "Collusion with Persistent Cost Shocks." *Econometrica*, Vol. 76 (2008), pp. 493–540.
- ATHEY, S., BAGWELL, K., and SANCHIRICO, C. "Collusion and Price Rigidity." *The Review of Economic Studies*, Vol. 71 (2004), pp. 317–349.
- AUSTIN, D.R. and BAKER, L.C. "Less Physician Practice Competition is Associated with Higher Prices Paid for Common Procedures." *Health Affairs*, Vol. 34 (2015), pp. 1753–1760.
- BAKER, J.B., FARRELL, J., and SHAPIRO, C. "Merger to Monopoly to Serve a Single Buyer: Comment." *Antitrust LJ*, Vol. 75 (2008), p. 637.
- BAKER, L.C., BUNDORF, M.K., ROYALTY, A.B., and LEVIN, Z. "Physician Practice Competition and Prices Paid by Private Insurers for Office Visits." *Jama*, Vol. 312 (2014), pp. 1653–1662.
- BERRY, S.T. "Estimating Discrete-Choice Models of Product Differentiation." *The RAND Journal of Economics*, (1994), pp. 242–262.
- BINGAMAN, A. "Recent Enforcement Actions By the Antitrust Division Against Trade Associations." 32nd Annual Symposium of the Trade Association and Antitrust Law Committee of the Bar Association of the District of Columbia, US Department of Justice, Antitrust Division, 1996.
- BITRAN, R., ESCOBAR, L., and GASSIBE, P. "After Chile's Health Reform: Increase in Coverage and Access, Decline in Hospitalization and Death Rates." *Health Affairs*, Vol. 29 (2010), pp. 2161–2170.
- BLAIR, R.D. and DePASQUALE, C. "Considerations of Countervailing Power." *Review of Industrial Organization*, Vol. 39 (2011), p. 137.
- BLAIR, R.D. and HERNDON, J.B. "Physician Cooperative Bargaining Ventures: An Economic Analysis." *Antitrust Law Journal*, Vol. 71 (2004), pp. 989–1016. <http://www.jstor.org/stable/40843610>.
- BRESNAHAN, T.F. "Competition and Collusion in the American Automobile Industry: The 1955 Price War." *The Journal of Industrial Economics*, Vol. 35 (1987), pp. 457–482.
- . "Bilateral Monopoly in Mergers." *Antitrust Law Journal*, Vol. 74 (2007), pp. 521–536.
- . "Bilateral Monopoly: Further Comment." *Antitrust LJ*, Vol. 75 (2008), p. 647.
- CARD, D. "Strikes and Wages: A Test of an Asymmetric Information Model." *The Quarterly Journal of Economics*, Vol. 105 (1990), pp. 625–659.
- CARNEVALI, F. "Social Capital and Trade Associations in America, c. 1860–1914: A Microhistory Approach." *The Economic History Review*, Vol. 64 (2011), pp. 905–928.
- CILIBERTO, F. and WILLIAMS, J.W. "Does Multimarket Contact Facilitate Tacit Collusion? Inference on Conduct Parameters in the Airline Industry." *The RAND Journal of Economics*, Vol. 45 (2014), pp. 764–791.
- CLARK, R. and HOUDE, J.F. "Collusion with Asymmetric Retailers: Evidence from a Gasoline Price-Fixing Case." *American Economic Journal: Microeconomics*, Vol. 5 (2013), pp. 97–123.
- Competition Commission of Singapore. "CCS Fines 16 Coach Operators and Association US\$ 1.69 Million For Price-Fixing." Press Release, 2009.
- CORTS, K.S. "Conduct Parameters and the Measurement of Market Power." *Journal of Econometrics*, Vol. 88 (1999), pp. 227–250.
- CRAMTON, P. and TRACY, J. "Unions, Bargaining and Strikes." In J.T. Addison, H.C. Lane, eds., *International Handbook of Trade and Unions*. Cheltenham, UK: Edward Elgar, 2003.
- DOJ and FTC. "Statement of Antitrust Enforcement Policy Regarding Accountable Care Organizations Participating in the Medicare Shared Savings Program; Notice, Part II." Federal Register, 2011.
- DONOVAN, W.J. "The Legality of Trade Associations." *Proceedings of the Academy of Political Science in the City of New York*, Vol. 11 (1926), pp. 19–26.
- DUNN, A. and SHAPIRO, A.H. "Do Physicians Possess Market Power?" *The Journal of Law & Economics*, Vol. 57 (2014), pp. 159–193.
- EIZENBERG, A. and SALVO, A. "The Rise of Fringe Competitors in the Wake of an Emerging Middle Class: An Empirical Analysis." *American Economic Journal: Applied Economics*, Vol. 7 (2015), pp. 85–122.

- FNE. "Requerimiento." *Rol C. No. 265-13. Fiscalía Nacional Económica*, 2013.
- FNE. "Observaciones a la Prueba." *Rol C No. 265-13*, 2014.
- FTC. "Spotlight on Trade Associations." <https://www.ftc.gov/tips-advice/competition-guidance/guide-antitrust-laws/dealings-competitors/spotlight-trade> (2018). Accessed: June 30, 2018.
- FULTON, B.D. "Health Care Market Concentration Trends in the United States: Evidence and Policy Responses." *Health Affairs*, Vol. 36 (2017), pp. 1530–1538.
- GANDHI, A., LU, Z., and SHI, X. "Demand Estimation with Scanner Data: Revisiting the Loss-Leader Hypothesis." *Working Paper*, 2014.
- GENESOVE, D. and MULLIN, W.P. "Testing Static Oligopoly Models: Conduct and Cost in the Sugar Industry, 1890–1914." *The RAND Journal of Economics*, (1998), pp. 355–377.
- . "Rules, Communication, and Collusion: Narrative Evidence from the Sugar Institute Case." *American Economic Review*, Vol. 91 (2001), pp. 379–398.
- GOWRISANKARAN, G., NEVO, A., and TOWN, R. "Mergers When Prices are Negotiated: Evidence from the Hospital Industry." *American Economic Review*, Vol. 105 (2015), pp. 172–203.
- GU, W. and KUHN, P. "A Theory of Holdouts in Wage Bargaining." *American Economic Review*, Vol. 88 (1998), pp. 428–449.
- HARRINGTON, J.E. "Heterogeneous Firms Can Always Collude on a Minimum Price." *Economics Letters*, Vol. 138 (2016), pp. 46–49.
- HARRINGTON, J.E. and SKRZYPACZ, A. "Private Monitoring and Communication in Cartels: Explaining Recent Collusive Practices." *American Economic Review*, Vol. 101 (2011), pp. 2425–49.
- HICKS, J. *The Theory of Wages*. New York: Macmillan, 1932.
- HO, K. and LEE, R.S. "Insurer Competition in Health Care Markets." *Econometrica*, Vol. 85 (2017), pp. 379–417.
- IGAMI, M. and SUGAYA, T. "Measuring the Incentive to Collude: The Vitamin Cartels, 1990–1999." Working Paper, 2017.
- INDH. "Informe Anual 2018: Situación de los Derechos Humanos en Chile." <https://bibliotecadigital.indh.cl/handle/123456789/1173> (2018). Accessed: 2019-07-24.
- KENNAN, J. "The Economics of Strikes." *Handbook of Labor Economics*, Vol. 2 (1986), pp. 1091–1137.
- KÜHN, K.U., MATUTES, C., and MOLDOVANU, B. "Fighting Collusion by Regulating Communication between Firms." *Economic Policy*, Vol. 16 (2001), pp. 169–204.
- KIRBY, A.J. "Trade Associations as Information Exchange Mechanisms." *The RAND Journal of Economics*, Vol. 19 (1988), pp. 138–146.
- LEE, R.S. and FONG, K. "Markov Perfect Network Formation: An Applied Framework for Bilateral Oligopoly and Bargaining in Buyer-Seller Networks." Working Paper, 2013.
- LEVENSTEIN, M.C. "Price Wars and the Stability of Collusion: A Study of the Pre-World War I Bromine Industry." *The Journal of Industrial Economics*, Vol. 45 (1997), pp. 117–137.
- LEVENSTEIN, M.C. and SUSLOW, V.Y. "What Determines Cartel Success?" *Journal of Economic Literature*, Vol. 44 (2006), pp. 43–95.
- . "Breaking Up Is Hard to Do: Determinants of Cartel Duration." *Journal of Law & Economics*, Vol. 54 (2011), pp. 455–462.
- MCGAHAN, A.M. "Cooperation in Prices and Capacities: Trade Associations in Brewing After Repeal." *The Journal of Law and Economics*, Vol. 38 (1995), pp. 521–559.
- MEIER, M., ALBERT, B., and MONAHAN, K. "Overview of FTC Actions in Health Care Services and Products." https://www.ftc.gov/system/files/attachments/competition-policy-guidance/overview_health_care_august_2018.pdf. Accessed 8/29/19
- MILLER, N.H. and WEINBERG, M.C. "Understanding the Price Effects of the MillerCoors Joint Venture." *Econometrica*, Vol. 85 (2017), pp. 1763–1791.
- Modern Healthcare. "FTC Keeps Wary Eye on Network Arrangements." <https://www.modernhealthcare.com/article/20090112/MODERNPHYSICIAN/301049974/ftc-keeps-wary-eye-on-network-arrangements>. Accessed August 29, 2019.
- MOTTA, M. *Competition Policy: Theory and Practice*. Cambridge, UK: Cambridge University Press, 2004.
- MULLAN, H. and TIMAN, N. "Strengthening Buyer Power as a Solution to Platform Market Power." Competition Policy International Antitrust Chronicle, 2018.
- NEVO, A. "Identification of the Oligopoly Solution Concept in a Differentiated-Products Industry." *Economics Letters*, Vol. 59 (1998), pp. 391–395. <https://EconPapers.repec.org/RePEc:eee:ecole:v:59:y:1998:i:3:p:391-395>.
- NEVO, A. "Measuring Market Power in the Ready-to-Eat Cereal Industry." *Econometrica*, Vol. 69 (2001), pp. 307–342.
- OECD. "Potential Pro-Competitive and Anti-Competitive Aspects of Trade/Business Associations." OECD, 2007.
- OLIPHANT, H. "Trade Associations and the Law." *Columbia Law Review*, Vol. 26 (1926), p. 381.
- PHELPS, C.E. and NEWHOUSE, J.P. "Coinsurance, the Price of Time, and the Demand for Medical Services." *The Review of Economics and Statistics*, Vol. 56 (1974), pp. 334–342.
- PORTER, R.H. "A Study of Cartel Stability: The Joint Executive Committee, 1880–1886." *The Bell Journal of Economics*, Vol. 14 (1983), pp. 301–314.
- QUAN, T.W. and WILLIAMS, K.R. "Product Variety, Across-Market Demand Heterogeneity, and the Value of Online Retail." *The RAND Journal of Economics*, Vol. 49 (2018), pp. 877–913.

- RÖLLER, L.H. and STEEN, F. "On the Workings of a Cartel: Evidence from the Norwegian Cement Industry." *American Economic Review*, Vol. 96 (2006), pp. 321–338.
- STAIGER, R.W. and WOLAK, F.A. "Collusive Pricing with Capacity Constraints in the Presence of Demand Uncertainty." *The RAND Journal of Economics*, Vol. 23 (1992), pp. 203–220.
- SULLIVAN, C. "The Ice Cream Split: Empirically Distinguishing Price and Product Space Collusion." Working Paper, 2017.
- SYMEONIDIS, G. *The Effects of Competition: Cartel Policy and the Evolution of Strategy and Structure in British Industry*. MIT Press, 2002.
- VIVES, X. "Trade Association Disclosure Rules, Incentives to Share Information, and Welfare." *The RAND Journal of Economics*, Vol. 21 (1990), pp. 409–430.
- WHINSTON, M.D. *Lectures on Antitrust Economics (Cairolis Lectures)*. Cambridge, MA: MIT Press, 2008.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table A1: Changes in list price and coinsurance rate in Chillán and neighboring cities

Figure A1: Locations and nests

Table A2: Nested Logit estimates: zero market shares correction

Table A3: Supplemental insurance coverage

Figure A2: Truncated distribution of monthly visits

Data S1